Homework 9 · Simple & compound interest · ANSWERS

OnMaths · Isabella · Year 8 Set 2 · Mr Ali · Calculator allowed · 10 marks

1. £250 is invested at 4% simple interest for 5 years. Work out the total amount after 5 years. **(2 marks)**

Answer: 250 × 0.04 = £10 a year; 10 × 5 = £50; total £300

2. Write down the multiplier for a 3.5% increase. **(1 mark)**

Answer: 1.035

3. £800 is invested at 5% compound interest per year. Work out the total after 3 years. Give your answer to the nearest penny. **(2 marks)**

Answer: $800 \times 1.05^3 = £926.10$

4. A phone is worth £600. It loses 25% of its value each year. Work out its value after 2 years. **(2 marks)**

Answer: $600 \times 0.75^2 = £337.50$

5. Mia invests £4000 for 4 years. Bank A pays 3% simple interest. Bank B pays 2.5% compound interest. Which bank gives Mia more money, and by how much? **(3 marks)**

Answer: A: £4480; B: $4000 \times 1.025^4 = £4415.25$; Bank A by £64.75

Show your working. Money answers to 2 decimal places.